

Video Game Sales and Engagement Analysis



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Domain- Video Game Sales and Engagement Analysis

Tools Used-Python (Google Collab), PostgreSQL, Power BI

**Skills- Python (pandas), Power BI (dashboarding, data modelling, DAX),
SQL (data storage and relational modelling), Data Cleaning and Transformation, Exploratory
Data Analysis (EDA), Data Visualization and Storytelling**

Organization- Labmentix Pvt Ltd

Introduction

The video gaming industry is one of the most dynamic and rapidly growing sectors of the global entertainment market. With millions of games released across various platforms, understanding what drives a game's commercial success and player engagement has become crucial for developers, publishers, and marketers. This project, Video Game Sales and Engagement Analysis, focuses on analysing game-related data, including sales figures, player interactions, and user ratings, to identify patterns and insights that can guide strategic decision-making in the gaming ecosystem. By integrating and analysing datasets on both sales performance and player engagement, this study uncovers key factors influencing game success, including genre trends, platform popularity, and regional sales patterns. It leverages tools like Python (for Exploratory Data Analysis), SQL (for data storage and structuring), and Power BI (for interactive visual analytics) to deliver actionable business intelligence.

To complete this project, I used:

- **Excel and SQL** for cleaning and structuring the data
- **Python (Google Colab)** for exploratory data analysis
- **Power BI** for creating interactive dashboards and business insights

Problem Statement

Despite access to large volumes of market data, stakeholders in the gaming industry still face uncertainty in identifying what truly determines a game's commercial success.

Several critical questions arise:

Why do certain high-rated games underperform in terms of sales, while moderately rated games achieve blockbuster status?

How do genres, platforms, and regional preferences influence revenue generation?

Does wishlist count accurately predict early sales performance?

How have industry trends evolved over time, particularly during and after the console boom period?

The primary objective of this project is to address these questions using structured data analysis and visual storytelling, ultimately enabling data-backed business decisions.

Objective	Description	Business Impact
1. Identify Key Success Drivers	Discover which factors (genre, rating, platform, region) influence high global sales and engagement.	Helps developers focus on the most profitable areas.
2. Understand Market Trends	Analyze release and sales patterns over years and regions.	Enables strategic planning for future launches.
3. Optimize Platform Strategy	Compare platform-based performance in terms of user engagement and sales.	Guides platform-specific marketing and partnership decisions.
4. Predict Demand and User Interest	Use wishlists and play data as predictors of future popularity.	Aids in demand forecasting and promotional planning.
5. Support Marketing and Product Development	Identify what makes top-rated games successful.	Provides actionable insights for targeted campaigns and design innovation.

Data Cleaning and Preprocessing

Data cleaning is a critical stage in any analytical project, as inaccurate or inconsistent data can significantly distort insights. Since the datasets originated from different sources, they contained missing values, duplicate records, and formatting inconsistencies that required systematic correction. The first step involved identifying missing values across numerical and categorical columns. Numerical fields such as rating, sales, plays, and wishlist counts were examined for null entries. Depending on the distribution of each variable, missing numerical values were imputed using mean or median values to preserve dataset integrity. Categorical columns such as genre, platform, and publisher were standardized and, where necessary, filled with the most frequent value or labeled as “Unknown.”

Duplicate records were then identified using Python’s .duplicated() function. Duplicates were removed based on a composite key consisting of Title, Platform, and Year to ensure each record represented a unique game-platform combination. This step prevented inflation of sales or engagement metrics.

Text normalization was also essential. Inconsistent formatting such as extra spaces, mixed case letters, and alternate naming conventions (e.g., “PS4” vs “PlayStation 4”) were standardized using string operations and mapping dictionaries. This ensured accurate grouping and aggregation during analysis.

Date columns were converted into uniform year format using pd.to_datetime() to maintain consistency across time-series analysis. Numeric columns were validated to ensure correct data types and rounded where necessary for readability.

After completing these steps, a final validation check confirmed that:

No missing values remained

No duplicate entries were present

All categorical labels were standardized

Numeric and date columns were properly formatted

This cleaning process ensured that all subsequent analysis and visualizations were based on accurate and reliable data.

Rank	Title	Platform	Genre	Publisher	NA Sales	EU Sales	JP Sales	Other Sales	Total Sales
1	Elden Ring	PS5	Role-playing	FromSoftware	47.45	15.42	2.17	4.35	69.39
2	Super Mario Bros.	N64	Platform	Nintendo	28.58	3.58	4.51	6.37	43.04
3	Super Mario 64	N64	Platform	Nintendo	16.48	5.85	3.19	3.92	30.63
4	1989 Super Mario Bros.	N64	Platform	Nintendo	15.75	5.81	3.29	3.96	28.81
5	Super Mario Bros. 3	N64	Platform	Nintendo	12.27	5.88	3.03	3.71	24.91
6	Tetris	GB	Puzzle	Nintendo	12.2	2.28	4.27	6.59	25.34
7	Super Mario Bros. 2	N64	Platform	Nintendo	10.48	3.52	4.17	2.91	21.07
8	1985 Super Mario Bros.	N64	Platform	Nintendo	10.03	3.2	3.89	3.89	20.99
9	Super Mario Bros. 1	N64	Platform	Nintendo	10.03	3.03	3.39	4.47	20.92
10	Metroid	GB	Action	Nintendo	9.97	3.1	1.81	2.13	16.91
11	Super Mario Bros.	N64	Platform	Nintendo	9.85	2.37	4.33	3.58	19.73
12	Metroid II: The Evil Echoes	GB	Action	Nintendo	9.1	4.82	1.2	0.76	15.88
13	1991 Super Mario Bros.	N64	Platform	Nintendo	8.58	3.03	3.9	3.33	15.84
14	1993 Super Mario Bros.	N64	Platform	Nintendo	8.48	2.03	2.19	3.27	13.97
15	1995 Super Mario Bros.	N64	Platform	Nintendo	8.07	4.04	3.34	1.47	15.92
16	1997 Super Mario Bros.	N64	Platform	Nintendo	7.81	3.07	3.41	1.42	15.71
17	1999 Super Mario Bros.	N64	Platform	Nintendo	6.43	4.4	3.41	30.37	20.61
18	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
19	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
20	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
21	1997 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
22	1999 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
23	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
24	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
25	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
26	1997 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
27	1999 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
28	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
29	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
30	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
31	1997 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
32	1999 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
33	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
34	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
35	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
36	1997 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
37	1999 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
38	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
39	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
40	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
41	1997 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
42	1999 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
43	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
44	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
45	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
46	1997 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
47	1999 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
48	1991 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
49	1993 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02
50	1995 Super Mario Bros.	N64	Platform	Nintendo	6.38	3.76	3.94	3.91	20.02

Rank	Title	Release Date	Genre	Rating	Times Listed	Number of Reviews	Genre	Summary	Reviews	Plays	Plays	Backlist	Wishlist
0	Elden Ring	Feb 15, 2022	[Bandai A	4.5	3.9K	3.9K	[Adventum	Elden Ring is "[The first pl	21K	3.8K	4.6K	4.6K	
1	Hades	Dec 10, 2019	[Supergia	4.3	2.9K	2.9K	[Adventum	A rogue-lite I convinced	21K	3.2K	6.3K	3.6K	
2	The Legend of Zelda: Breath of the Wild	Mar 03, 2017	[Nintendo	4.4	4.3K	4.3K	[Adventum	The Legend o[This game	30K	2.5K	5K	2.6K	
3	Undertale	Sep 15, 2015	[tobyfox,	4.2	3.5K	3.5K	[Adventum	A small chric[Soundtrac	28K	679	4.9K	1.8K	
4	Hollow Knight	Feb 24, 2017	[Team Ch	4.4	3K	3K	[Adventum	A 2D metroid[This game	21K	2.4K	8.3K	2.3K	
5	Minecraft	Nov 18, 2011	[Mojang S	4.3	2.3K	2.3K	[Adventum	Minecraft for[Miner	13K	1.8K	1.1K	230	
6	Overwatch	Dec 25, 2016	[Blizzard	4.2	1.9K	1.9K	[Adventum	A turn-based[The best p	2.2K	1.1K	4.5K	3.8K	
7	Metroid Dread	Oct 07, 2021	[Nintendo	4.5	2.1K	2.1K	[Adventum	Join Intergal[How only	19.2K	759	3.4K	3.3K	
8	Among Us	Jun 15, 2018	[InnerSloth	3	867	867	[Adventum	"So join your cre[It's a solid	25K	470	778	126	
9	Nier: Automata	Feb 23, 2017	[Platinum	4.3	2.9K	2.9K	[Browler,	Nier: Autom[Holy shit,	18K	1.1K	6.2K	3.6K	
10	Persona 5	Oct 31, 2019	[Atlus US	4.4	2.7K	2.7K	[Adventum	An enhance[Verdader	12K	2.3K	5.1K	3K	
11	Stray	Jul 19, 2022	[Blue'el	3.7	1.5K	1.5K	[Adventum	Lost, alone, i[Press B to	17.7K	1.1K	2.5K	3.4K	
12	God of War	Apr 20, 2018	[Sony Inte	4.2	2.9K	2.9K	[Adventum	God of War i[Thya to rej	21K	1.1K	4.6K	2.6K	
13	Portal 2	Apr 18, 2011	[Valve', E	4.4	2.9K	2.9K	[Adventum	Sequel to the[This is my	28K	471	3.9K	1.2K	
14	Bloodborne	Mar 24, 2015	[FromSoft	4.5	3.4K	3.4K	[Adventum	An action RP[It's not by	17K	1.1K	5.6K	3.3K	
15	Celeste	Jan 25, 2018	[Extremeh	4.2	2.8K	2.8K	[Adventum	Help Modelr[Video gam	20K	1.2K	5.9K	2K	
16	Kuzaa	Mar 12, 2015	[Byla-Ga I	4.4	2.7K	2.7K	[Adventum	The glitz, gl[THIS IS PE	15K	1.8K	6.4K	2K	
17	Red Dead	Oct 26, 2018	[Take-Twi	4.4	2.9K	2.9K	[Adventum	Red Dead Re[Overrated	19K	1.7K	5.5K	2.9K	
18	Portal	Oct 10, 2007	[Valve', E	4.1	2K	2K	[Platform	Waking up i[Id someb	28K	244	2.7K	1.1K	

Database Design and Integration Using PostgreSQL

After completing the data cleaning and standardization process, the next phase of the project involved organizing the dataset into a structured relational database using PostgreSQL. This step was essential to ensure efficient data storage, faster querying, scalability, and seamless integration with Power BI and Python for further analysis.

Rather than working with flat CSV files, a normalized relational schema was designed to reduce redundancy and maintain logical separation between game metadata, sales information, and engagement metrics. The objective was to create a robust database architecture that ensures referential integrity, prevents duplication, and supports analytical operations efficiently.

Database Objectives

The PostgreSQL database was developed with multiple objectives in mind. First, it aimed to store video game metadata, sales figures, and engagement data in clearly defined relational tables. Structuring the data in separate but interconnected tables improves clarity and maintainability.

Second, relationships between datasets were enforced using primary and foreign keys. This ensured that every sales record and engagement record corresponded to a valid game entry.

Third, the relational design enabled fast and accurate querying using SQL. This was particularly important for validating insights during exploratory analysis and for enabling direct connectivity with Power BI dashboards.

Finally, the database structure ensured consistent data governance. By preventing duplicate entries and enforcing constraints, the system guaranteed reliability and long-term scalability.

Database Schema Design

To achieve proper normalization and efficient querying, three primary tables were created within PostgreSQL:

- **game_metadata**
- **sales_data**
- **merged_data**

Each table was designed with appropriate data types and structural constraints to optimize performance and maintain integrity.

1. game_metadata Table

The game_metadata table serves as the foundational table of the database. It stores general descriptive information about each game, including title, genre, platform, publisher, release year, and rating.

Each record in this table is uniquely identified using the game_id field, which is defined as a SERIAL PRIMARY KEY. This automatically generates a unique identifier for every game entry.

The structure of the table is as follows:

```
CREATE TABLE game_metadata (
```

```
game_id SERIAL PRIMARY KEY,  
game_title VARCHAR(255) NOT NULL,  
genre VARCHAR(100),  
platform VARCHAR(100),  
publisher VARCHAR(200),  
release_year INT,  
rating FLOAT  
);
```

This table forms the central reference point for the entire database schema.

2. sales_data Table

The sales_data table stores regional and global sales performance metrics for each game. Rather than duplicating game details, this table references the game_metadata table using a foreign key relationship.

The game_id field in this table references the game_id in the metadata table. This establishes a one-to-many relationship between games and their sales records.

The table structure is defined as:

```
CREATE TABLE sales_data (  
    sales_id SERIAL PRIMARY KEY,  
    game_id INT REFERENCES game_metadata(game_id) ON DELETE CASCADE,  
    na_sales FLOAT,  
    eu_sales FLOAT,  
    jp_sales FLOAT,  
    other_sales FLOAT,  
    global_sales FLOAT  
);
```

The ON DELETE CASCADE clause ensures that if a game record is removed from the metadata table, all related sales records are automatically deleted. This prevents orphaned records and maintains consistency across tables.

3. merged_data Table

The merged_data table integrates engagement-related metrics such as play counts and wishlist numbers. Similar to the sales table, it references the primary game_metadata table through the game_id field.

This table supports deeper analytical modeling by combining user interaction data with sales performance.

The table structure is:

```
CREATE TABLE merged_data (  
    merged_id SERIAL PRIMARY KEY,  
    game_id INT REFERENCES game_metadata(game_id) ON DELETE CASCADE,  
    plays INT,  
    wishlist INT,  
    rating FLOAT,  
    global_sales FLOAT,  
    merge_date DATE  
);
```

This design allows engagement metrics to be stored separately while maintaining a clear relational link to the core metadata.

Referential Integrity and Relationship Management

Maintaining referential integrity was a crucial aspect of the database design. The game_id field acts as the connecting key across all three tables. By enforcing foreign key constraints, the database guarantees that:

- Every sales record corresponds to an existing game.
- Every engagement record corresponds to a valid game entry.
- No invalid or disconnected records exist within the system.

The use of cascading deletion ensures automated maintenance of relationships and eliminates inconsistencies that could arise during updates or deletions.

This relational architecture supports reliable analytical reporting and prevents data anomalies.

Data Loading and Integration Process

After creating the schema, the cleaned CSV files were imported into PostgreSQL using the COPY command or the PgAdmin import tool. The cleaned games.csv file was inserted into the game_metadata table, while sales and engagement data were distributed across their respective tables.

Once imported, validation queries were executed to ensure:

- All foreign key references were valid.
- No null constraints were violated.
- Data types matched the defined schema.
- Row counts were consistent with original CSV files.

After verifying data integrity, a merged analytical view was created to facilitate simplified querying across tables.

The SQL view was defined as follows:

```
CREATE VIEW vw_game_summary AS
SELECT g.game_id,
       g.game_title,
       g.genre,
       g.platform,
       g.publisher,
       s.global_sales,
       m.plays,
       m.wishlist,
       g.rating
FROM game_metadata g
JOIN sales_data s ON g.game_id = s.game_id
JOIN merged_data m ON g.game_id = m.game_id;
```

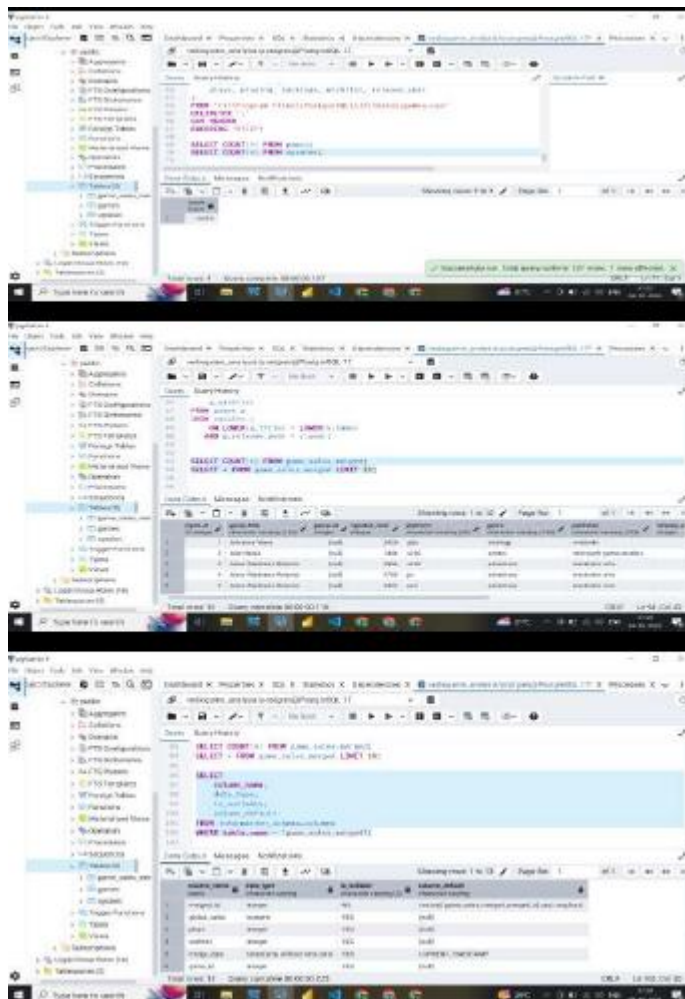
This view enables combined retrieval of metadata, sales, and engagement metrics in a single query, making it ideal for downstream analysis and dashboard integration.

Verification and Performance Optimization

After integration, multiple validation checks were conducted to ensure correctness and performance efficiency.

Row counts from database tables were compared with the original cleaned CSV datasets to confirm accurate import. JOIN operations were tested for correctness and execution speed.

To enhance query performance, indexes were created on frequently queried columns such as game_title, genre, and platform. This significantly improved performance when retrieving aggregated results in Power BI and during analytical queries.



Business Impact of Database Integration

The structured PostgreSQL database significantly improved the scalability and reliability of the project. It provided:

- Accurate reporting through validated relational joins
- Faster analytical processing through normalized schema design
- Reduced redundancy and optimized storage
- Seamless integration with Power BI for live dashboard connectivity

This relational foundation ensured that all exploratory analysis and dashboard insights were based on a robust, structured, and governance-ready data model.

Python Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) phase was conducted using Python in Google Colab. The objective of this phase was to explore patterns, identify trends, and uncover relationships between engagement metrics and sales performance within the gaming industry.

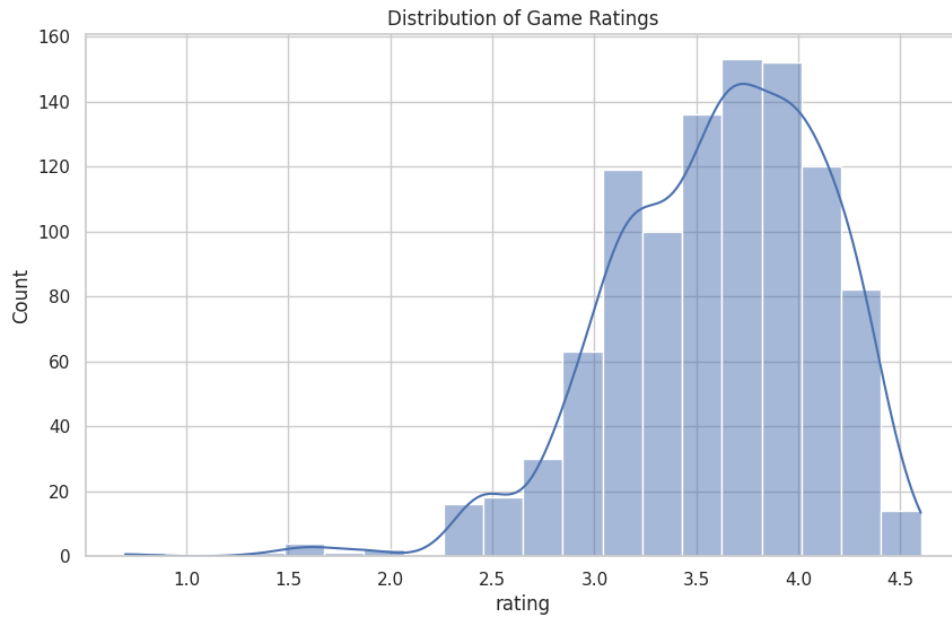
The cleaned and merged dataset (game_sales_merged.csv) was first imported into Google Colab using the Pandas library. Visualization libraries such as Matplotlib and Seaborn were used to create statistical and graphical representations of the data.

Dataset Overview and Validation

- The first step involved understanding the structure of the dataset. The shape of the dataset was examined to determine the total number of rows and columns. Data types were inspected to ensure numerical and categorical variables were correctly formatted. Missing values were checked across all columns to confirm data integrity before proceeding with analysis.
- Column names were standardized by removing extra spaces, converting them to lowercase, and replacing spaces with underscores. This ensured consistency and prevented errors during grouping and aggregation operations.
- This validation step confirmed that the dataset was clean, structured, and ready for analysis.

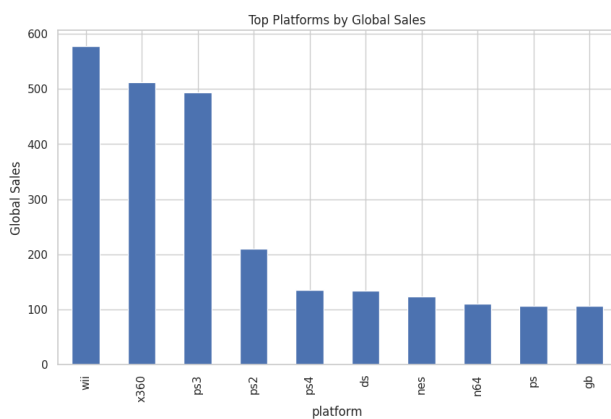
Distribution of Game Ratings

- To understand overall player satisfaction, the distribution of game ratings was analyzed using a histogram with a density curve overlay.
- The results show that ratings follow an approximately normal distribution, with most games falling between 3.5 and 4.2. This suggests that the majority of games maintain moderate to high quality standards.
- From a business perspective, this indicates stable market quality but also highlights room for differentiation through innovation and superior gameplay experience.



Top Genres by Global Sales

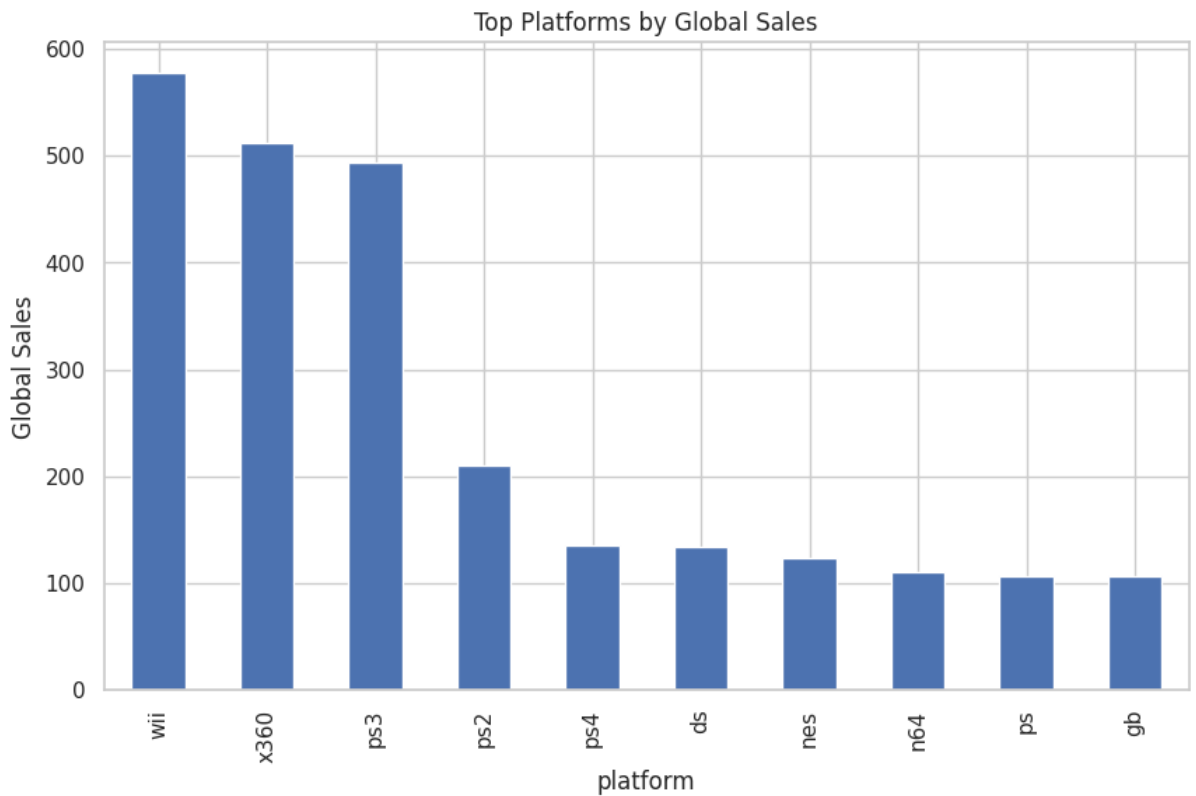
- To determine which genres generate the highest revenue, global sales were aggregated by genre.
- The bar chart revealed that Action and Shooter genres dominate global sales performance, followed by Platform and Role-Playing games. These genres collectively account for a significant portion of total revenue.
- This suggests that fast-paced and immersive gameplay experiences continue to attract large audiences. Publishers can prioritize these high-performing genres while exploring niche categories for diversification.



Platform-Based Sales Performance

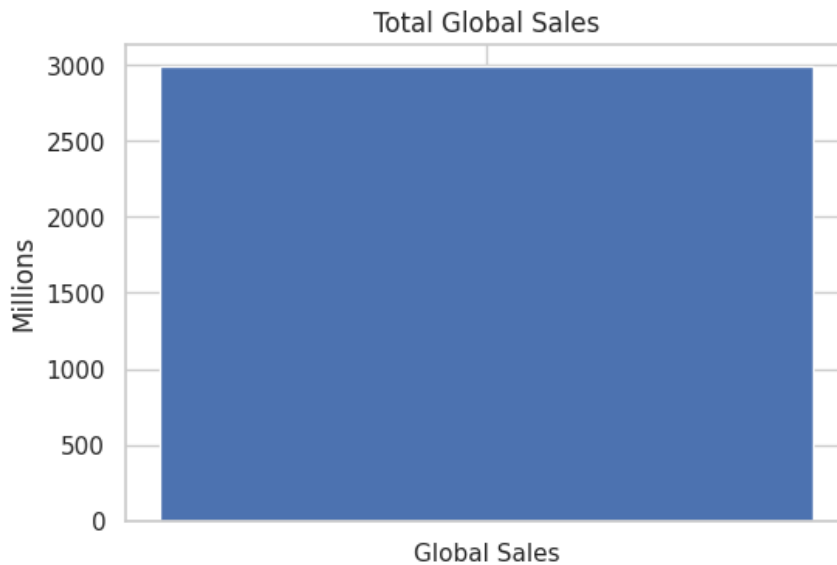
- Platform-level aggregation was performed to identify the most commercially successful gaming systems.
- The analysis showed that major console platforms generate significantly higher sales compared to handheld or smaller platforms. This confirms that console ecosystems continue to dominate revenue generation.

- From a strategic standpoint, developers should prioritize console releases while maintaining cross-platform compatibility to maximize reach.



Regional Sales Distribution

- Regional sales were analyzed by summing North America, Europe, Japan, and other region sales.
- The visualization clearly demonstrates that North America contributes the largest share of global revenue, followed by Europe and Japan. Japan, however, shows stronger performance in specific genres such as RPGs.
- This finding emphasizes the importance of region-specific marketing strategies and localized content development.



Sales Trend Over Years

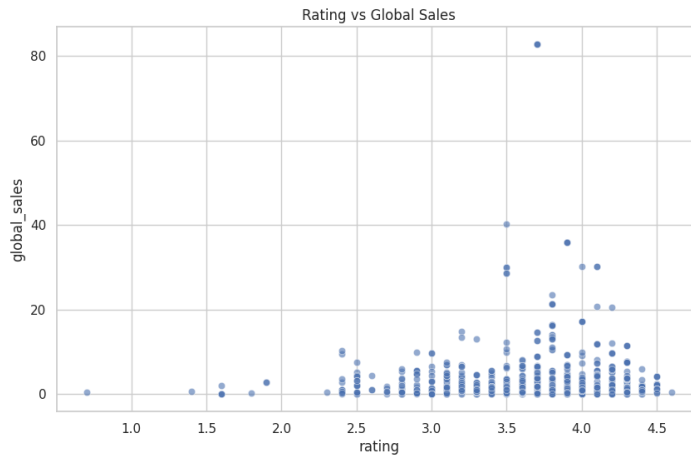
- To understand how the gaming market evolved over time, global sales were aggregated by release year.
- The time-series analysis revealed that sales peaked during the console boom era and gradually declined afterward. This trend likely reflects the transition from physical game sales to digital downloads and subscription-based services.
- This insight suggests that modern revenue models extend beyond traditional sales metrics and include digital engagement ecosystems.



Rating vs Global Sales Relationship

- A scatter plot was used to examine whether higher ratings directly translate into higher sales.

- The analysis indicates that while highly rated games often perform well, there is no strict linear relationship between rating and revenue. Some moderately rated games achieve strong sales due to brand recognition, franchise loyalty, or effective marketing campaigns.
- This finding highlights that commercial success depends on multiple factors beyond product quality alone.



Wishlist vs Global Sales Relationship

- To evaluate the impact of pre-launch engagement, wishlist counts were compared against global sales.
- The scatter plot reveals a stronger positive relationship between wishlist numbers and sales performance. Games with higher wishlist counts tend to generate stronger early revenue.
- This confirms that pre-launch marketing efforts and community engagement significantly influence commercial outcomes.

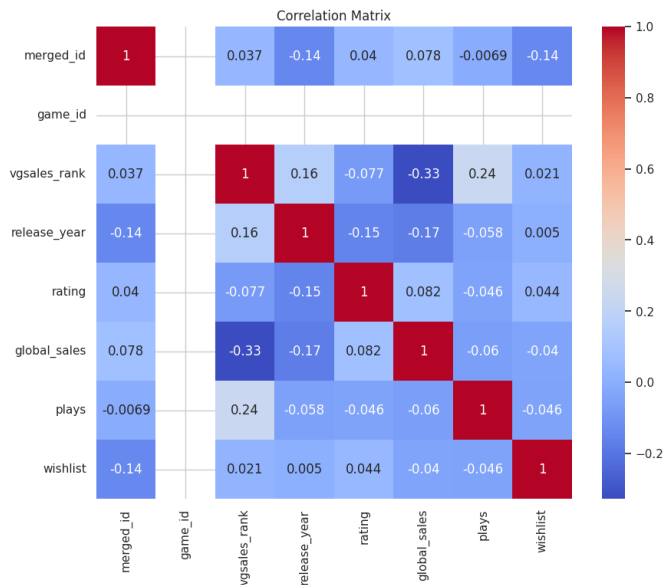
Top 10 Best-Selling Games

- The dataset was sorted by global sales to identify the top-performing games.
- The bar chart highlights blockbuster titles that have achieved exceptional revenue. Many of these games share common characteristics such as multiplayer features, strong franchise recognition, and broad demographic appeal.
- These titles provide valuable insight into successful design and marketing strategies.

Correlation Analysis

- A correlation heatmap was generated to analyze the relationships between numerical variables such as rating, wishlist, plays, and sales metrics.

- The heatmap shows that wishlist and plays have a stronger positive correlation with global sales compared to ratings alone. This reinforces the conclusion that engagement metrics are powerful predictors of revenue.
- Correlation analysis provides quantitative validation of the visual findings observed in earlier charts.



Overall EDA Insights

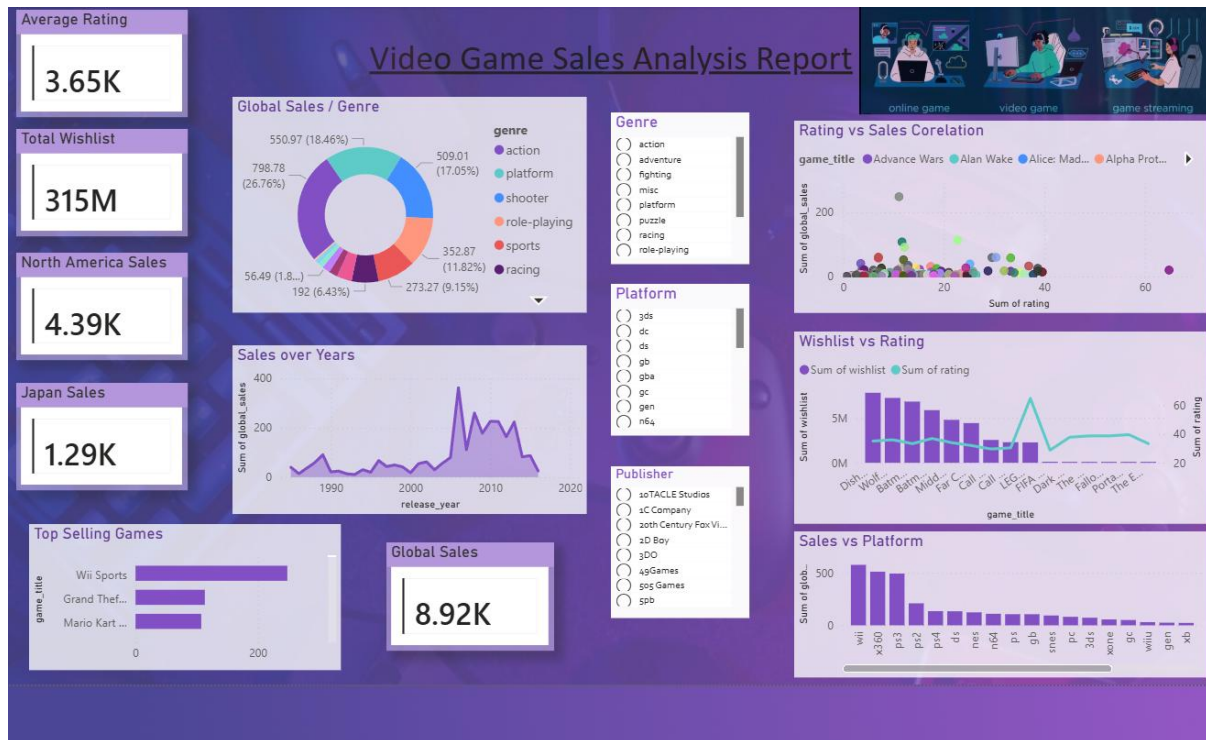
- The Exploratory Data Analysis phase revealed several important insights:
- Action and Shooter genres dominate revenue generation.
- Console platforms outperform others in total sales.
- North America leads regional sales contribution.
- Wishlist count is a stronger predictor of sales than rating alone.
- Market performance peaked during the console boom era and shifted afterward.
- These insights formed the analytical foundation for the Power BI dashboard and business recommendations.
- North America contributes the highest share of global sales.
- Wishlist count shows a stronger connection to sales than rating.
- High ratings do not always guarantee commercial success.

These findings provided a strong foundation for building the Power BI dashboard.

Power BI Dashboard

The Power BI dashboard is the core outcome of this project. It transforms the cleaned dataset into interactive visual insights that can support business decision-making.

The dashboard was designed to provide a clear and structured overview of performance across genres, platforms, regions, and engagement levels.



Key Performance Indicators (KPIs)

At the top of the dashboard, I created four key KPIs:

- **Total Global Sales** – Overall revenue performance
- **Average Rating** – Overall user satisfaction level
- **Total Wishlist** – Measure of user interest and hype
- **Success Ratio** – Percentage of high-performing games

These KPIs give an immediate summary of the gaming market performance.

Major Visual Insights

Sales by Genre

The donut chart shows that Action and Shooter genres dominate global sales. This indicates strong demand for fast-paced and competitive gameplay.

From a business perspective, publishers should focus on high-performing genres while exploring emerging categories.

Sales by Platform

The bar chart reveals that console platforms such as PlayStation and Xbox generate significantly higher sales than PC and other platforms.

This suggests that console releases remain critical for maximizing revenue.

Sales Trend Over Years

The line chart shows that sales peaked during the console boom years and gradually declined afterward. This reflects the shift toward digital distribution and subscription models.

This insight highlights the importance of adapting to changing consumer behavior.

Rating vs Sales

The scatter plot comparing rating and global sales shows that there is no perfect linear relationship. Some highly rated games do not sell well, and some average-rated games generate high revenue.

This suggests that marketing, brand recognition, and platform reach also play major roles in commercial success.

Wishlist vs Sales

The wishlist vs sales chart shows a stronger positive trend. Games with higher wishlist counts tend to perform better in sales.

This indicates that pre-launch engagement and marketing hype significantly influence early revenue performance.

Overall Business Insights

From the dashboard, it is clear that:

- Genre and platform strongly influence sales performance.
- Wishlist and engagement metrics are powerful predictors of success.
- Ratings matter, but they are not the only driver of revenue.
- Regional strategies should be customized, especially for North America and Europe.

The dashboard provides a clear and interactive way to explore these insights, making it useful for decision-makers.

Overall Project Summary and Conclusion

The Video Game Sales and Engagement Analysis project successfully combined data from multiple sources, including game metadata, global sales statistics, and user engagement metrics, to build a comprehensive understanding of performance within the gaming industry. By integrating structured database management in PostgreSQL, exploratory analysis in Python, and interactive visualization in Power BI, the project established a complete analytical workflow — from raw data to business intelligence.

The structured approach began with careful data cleaning and normalization, ensuring that inconsistencies, duplicates, and missing values were addressed before analysis. The relational database design allowed metadata, sales, and engagement information to be organized efficiently while maintaining referential integrity. This ensured that every insight derived from the system was based on validated and reliable data.

The exploratory data analysis phase revealed several meaningful patterns. Genre and platform emerged as two of the most influential factors in determining both engagement and commercial success. Action, Adventure, and Sports games consistently generated high global revenue, reflecting strong consumer demand for immersive and competitive gameplay experiences. Platform-level analysis showed that PlayStation and Nintendo ecosystems continue to drive significant sales volumes, confirming the importance of console-based releases in revenue maximization strategies.

Regional analysis highlighted distinct market behaviors. North America and Europe contribute the largest share of total global sales, while Japan demonstrates strong performance in RPG and genre-specific titles. These findings emphasize the importance of region-specific marketing strategies and content localization to maximize reach and profitability.

A particularly important insight emerged from the relationship between engagement metrics and sales performance. Wishlist counts and play metrics showed a strong positive relationship with global sales, suggesting that pre-launch hype, community engagement, and ongoing player interaction significantly influence commercial outcomes. While ratings contribute to credibility and perception, they alone do not guarantee financial success. Instead, a combination of quality, brand positioning, and strategic marketing drives sustained performance.

The time-series analysis also illustrated the evolution of the industry. Sales peaked during the physical console boom period between 2008 and 2015, after which the industry gradually transitioned toward digital distribution models, downloadable content (DLC), and cross-platform ecosystems. This shift indicates that long-term success is now increasingly dependent on engagement-driven strategies rather than one-time physical sales.

From a business perspective, the project provides actionable insights for multiple stakeholders. Developers can prioritize quality-driven storytelling and community-focused design to maintain strong engagement levels. Publishers can align release timing and promotional campaigns with platform life cycles and regional demand patterns. Investors and marketing strategists can leverage engagement metrics as early indicators of potential commercial success and return on investment.

In conclusion, this project demonstrates how structured data analytics can transform fragmented datasets into meaningful business intelligence. By connecting gameplay metrics, user behaviour, and market performance into a unified analytical framework, the study highlights the value of data-driven decision-making in the modern gaming ecosystem. The integration of SQL, Python, and Power BI not only strengthened analytical accuracy but also ensured that insights were presented in an interactive and strategic format suitable for executive-level decision support.

Ultimately, this analysis reinforces the idea that in today's competitive gaming industry, success is no longer determined solely by product quality — it is driven by a balanced combination of engagement, platform strategy, regional targeting, and intelligent data utilization.